NONLINEAR OPTIMIZATION MODEL FOR A CAMELID FIBER PROCESSING PLANT IN LA PAZ

I. INTRODUCTION

YACANA is a state-owned enterprise created by Supreme Decree No. 1979 of April 16, 2015. It has a social purpose, legal personality, and indefinite duration, and operates under the authority of the Minister of Productive Development and Plural Economy, who is responsible for sectoral policy.

The main business activity of the State-Owned Enterprise “YACANA” is the supply of raw materials, production, industrialization, and commercialization of products that form part of the Camelid Production Complex.

In this context, the company has industrial lines for processing South American camelid fiber, mainly alpaca and llama fiber, into washed fiber, tops (pre-carded fiber), yarn, and fabrics, with dyeing systems and the application of patterns and motifs.

At present, the company focuses on the preprocessing of camelid fiber and the production of tops (pre-carded fiber) and yarn. It also sells services for yarn production, yarn dyeing, and fabric dyeing, although the latter is less frequent. Tops are mainly exported regularly to England and China, where the company has stable clients willing to purchase the quantities it can produce. Although this product has lower added value, it offers greater sales predictability and generates foreign currency. Yarn, which has greater added value, is sold in the domestic market to different buyers, while services are likewise provided to national producers.

II. OBJECTIVE

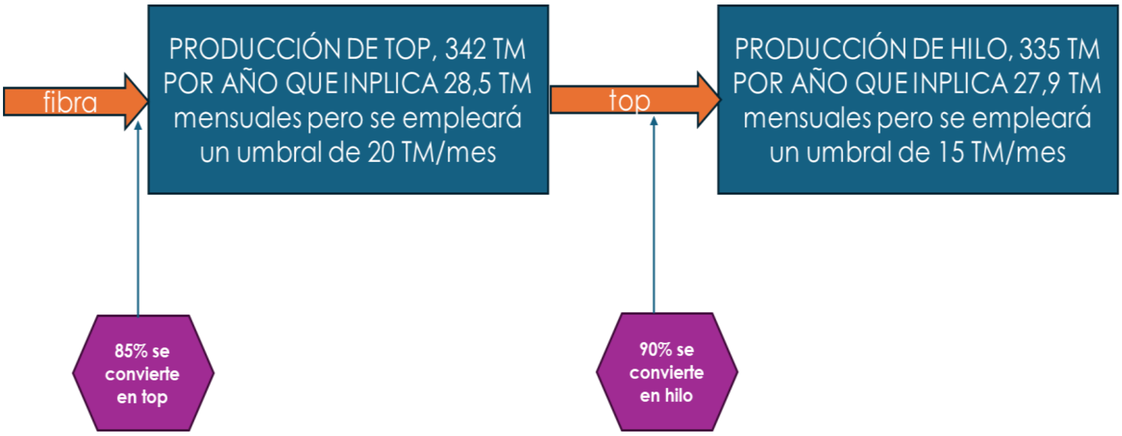
This paper presents the application of nonlinear programming to optimize a product mix between yarn and tops, which are the products currently manufactured by the company. The model considers an injection of working capital intended to maximize profit generation and minimize sales-related risk, under the conditions described above.

III. MODEL DEVELOPMENT

As indicated in the previous section, the model is aimed at identifying the best supply combination of alpaca tops and yarn, according to the current share of camelid fiber products to which the company has access.

The following figure summarizes the production capacities for tops and yarn, considering that tops are not only a finished product ready for sale, but also a necessary stage in yarn production (work in process).

Figure 1. Production capacities and relationships for tops and yarn



Source: Prepared by the author based on company data.

The prices of alpaca fiber (raw material) and of the finished products placed in the company's warehouses, at the time of writing this document, are shown below:

Figure 2. Prices of raw material and finished products

Precio de la fibra de alpaca por kg	180,00	Se adquiere de proveedores
top de alpaca por kg	280,00	Destinado a la exportación
Hilo de alpaca por kg	430,00	Destinado al mercado interno

Source: Prepared by the author based on company data.

The products manufactured are described in greater detail below, with regard to their commercial conditions.

Figure 3. Manufactured products considered in the model

Source: YACANA State-Owned Enterprise.

Alpaca carded fiber top product

This product is specifically intended for export to secured international clients (England and China), who may pay a 20% advance depending on the contracts signed, although the remaining payment is made within three months for each shipment, in foreign currency, mainly U.S. dollars. For this reason, a market risk of 5% is assumed. It is a product with low added value.

Alpaca yarn fiber product

This product is mainly intended for the domestic market; however, it still requires market consolidation and improvement of yarn quality according to client requirements. Payment is assumed to be received in the month in which the sale is made, in local currency (Bolivianos). It is a product with higher added value. A market risk of 30% is assumed.

Objective of the model

The model seeks to maximize profit, understood as the difference between sales and the cost of fiber, assuming that the remaining costs are generally not highly variable. This is achieved through a combination of quantities of tops sold and yarn sold, which the model must determine: the quantity of tops for sale and the quantity of yarn for sale, considering that part of the tops produced is allocated to yarn production according to its respective yield factor.

Main constraints

The capacities of the production lines must not be exceeded, considering the mass-balance and technological constraints of the industrial plant.

The model begins with an additional budget of two million Bolivianos. The monthly purchase of fiber must not exceed the available sources of funds, which begin with this fund and are then supplemented by payments from sales according to the payment terms of each product.

The product mix for sale must not exceed a risk level established as acceptable.

Mathematical model

Notation

<u>Variables</u>

x_i Quantity of tops sold in month i, in kg

y_i Quantity of yarn sold in month i, in kg

<u>Constants</u>

Prices

p_f Price of alpaca fiber in Bs per kg, 180

p_t Price of alpaca top in Bs per kg, 280

p_h Price of alpaca yarn in Bs per kg, 430

Derived results

Sales share

w_t Percentage share of top sales, calculated as the quantity sold over total sales.

w_h Percentage share of yarn sales, calculated as the quantity sold over total sales.

Capacities

c_t Top production capacity (20,000 kg per month)

c_h Yarn production capacity (15,000 kg per month)

Yields

r_ft Yield of fiber converted into tops: 85%

r_th Yield of tops converted into yarn: 90%

Risks

R_t Market risk of tops, assumed at 5%

R_h Market risk of yarn, assumed at 30%

Financing amount

M Financing amount in Bolivianos: 2,000,000

A business risk of 10%, calculated according to a portfolio logic, must not be exceeded.

Model: equations

Objective function (maximize total profits, that is, sales minus fiber purchases)

$$\boldsymbol{Max\ Z} = \sum_1^{12}((\boldsymbol{x_i p_x} + \boldsymbol{y_i p_y}) - ((\boldsymbol{x_i} + \boldsymbol{y_i} / \boldsymbol{r_{th}}) / \boldsymbol{r_{ft}})^* \boldsymbol{p_f})$$

Constraints

Fiber purchases must not exceed the financing capacity in the specific period.

Period 1:

$((\boldsymbol{x_1} + \boldsymbol{y_1}/\ \boldsymbol{r_{th}})/\ \boldsymbol{r_{ft}})^{*}\ \boldsymbol{p_f} \leq M + \mathbf{0{,}2} * \boldsymbol{x_i p_x}$

Period finaning availability

$s_1 = M + 0{,}2 * \ \boldsymbol{x_1 p_x} - ((\boldsymbol{x_1} + \boldsymbol{y_1}/\ r_{th})/\ r_{ft}\boldsymbol{)}$

Periods 2 and 3:

$((\boldsymbol{x_i} + \boldsymbol{y_i}/\ \boldsymbol{r_{th}})/\ \boldsymbol{r_{ft}})^{*}\ \boldsymbol{p_f} \leq s_{i-1} + y_{i-1} * p_y + 0{,}2 * \boldsymbol{x_i p_x}$

Period finaning availability

$s_i = \ s_{i-1} + y_{i-1} * p_y + 0{,}2 * \boldsymbol{x_i p_x}$ - $((\boldsymbol{x_i} + \boldsymbol{y_i}/\ r_{th})/\ r_{ft})^{*}\ p_f$

Fo the periods 4 to n:

$((\boldsymbol{x_i} + \boldsymbol{y_i}/\ \boldsymbol{r_{th}})/\ \boldsymbol{r_{ft}})^{*}\ \boldsymbol{p_f} \leq s_{i-1} + y_{i-1} * p_y + 0{,}2 * \boldsymbol{x_i p_x} + \mathbf{0}{,}8 * \boldsymbol{x_{i-3} p_x}$

Saldo del periodo

$s_i = s_{i-1} + y_{i-1} * p_y + 0{,}2 * \boldsymbol{x_i p_x} + 0{,}8 * \boldsymbol{x_{i-3} p_x}$ - $((\boldsymbol{x_i} + \boldsymbol{y_i}/\ r_{th})/\ r_{ft})^{*}\ p_f$

The output of tops and yards must be under the real lines capacities:

Tops $(\boldsymbol{x_i} + \boldsymbol{y_i}/\ r_{th}) \leq 20.000$

yarn $\boldsymbol{y_i} \leq 15.000$

The risk of business must be under 10%

$\sqrt{(\boldsymbol{w_t^2} * \boldsymbol{R_t^2}} + \boldsymbol{w_h^2} * \boldsymbol{R_h^2}\ ) \leq \ \ \boldsymbol{0{,}10}$

The Excel worksheet developed for applying Solver is shown below.

Table 1. Excel worksheet used to apply Solver for the implementation of the model described (variables, intermediate and final products, and production-line capacity constraints)

Variables			mes 1	mes 2	mes 3	mes 4	mes 5	mes 6	mes 7	mes 8	mes 9	mes 10	mes 11	mes 12
	Venta de tops en kg		8.865,74	5.014,81	2.836,58	1.604,48	9.710,90	10.472,39	8.740,22	6.537,00	13.340,15	15.040,77	15.040,77	15.040,77
	Venta de hilos en kg		2.630,88	1.488,13	841,75	476,13	2.881,68	3.107,65	2.593,63	1.939,84	3.958,65	4.463,31	4.463,31	4.463,31
	producción de top		11.788,94	6.668,29	3.771,85	2.133,51	12.912,77	13.925,34	11.622,03	8.692,38	17.738,65	20.000,00	20.000,00	20.000,00
	producción de hilo		2.630,88	1.488,13	841,75	476,13	2.881,68	3.107,65	2.593,63	1.939,84	3.958,65	4.463,31	4.463,31	4.463,31
Capacidades														
	Producción de tops	20	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00	20.000,00
	Producción de hilos	15	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00	15.000,00

This first section includes the decision variables in the blue block, considering the intermediate transformations involved in top production, according to raw material yields, and yarn production, according to the yields obtained from the use of tops. The block also presents the capacities for each line and time period, which constitute the limits that must not be exceeded, in accordance with the model explained above.

The following table presents the intermediate calculations related to fiber purchases, considering that the company has initial capital corresponding to 20% of top sales (an advance payment made by importers), to which the Bs 2 million injection is added.

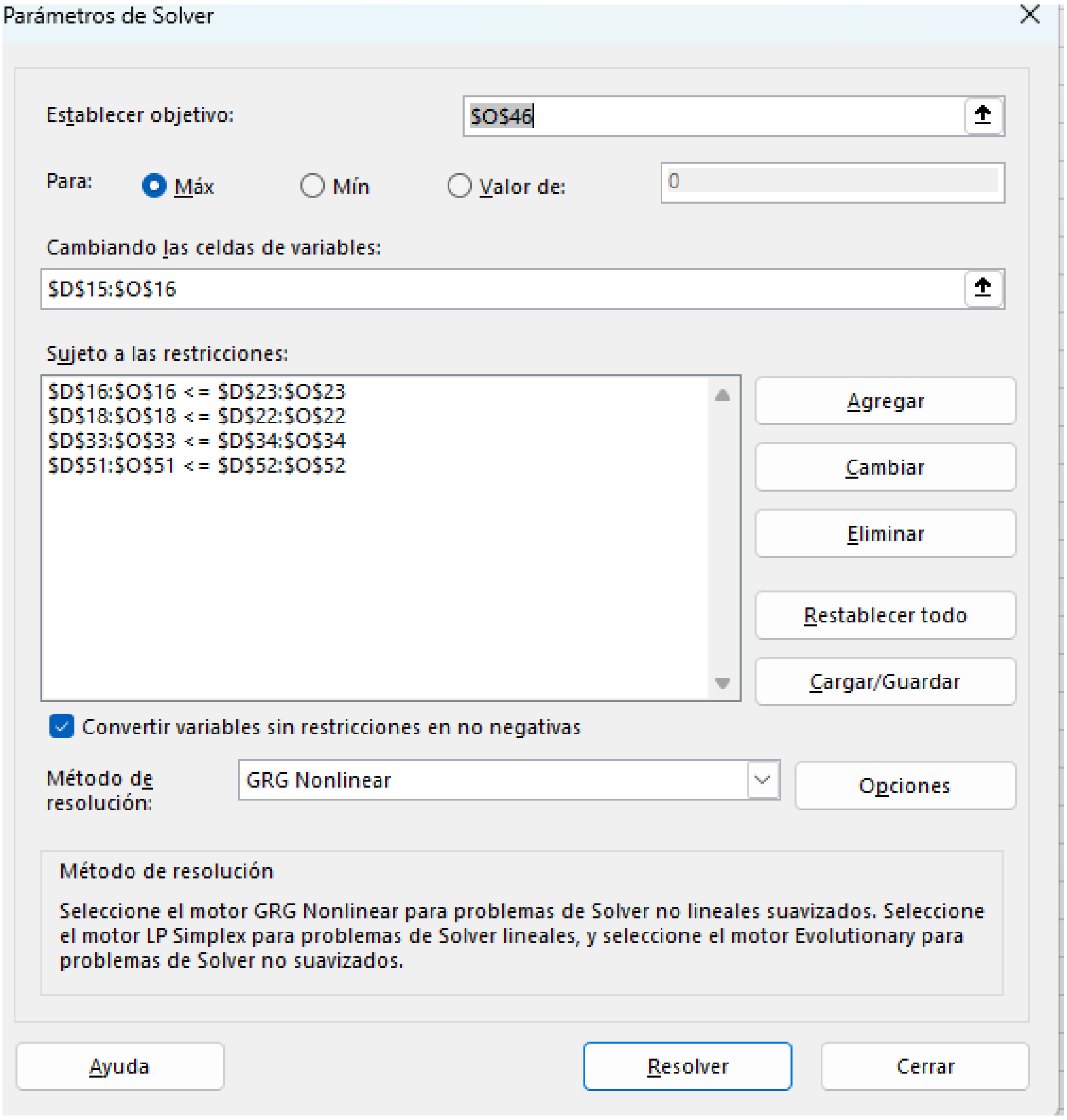
Table 2. Excel worksheet used to apply Excel Solver for the implementation of the model described (constraints)

			1	2	3	4	5	6	7	8	9	10	11	12
			mes 1	**mes 2**	**mes 3**	**mes 4**	**mes 5**	**mes 6**	**mes 7**	**mes 8**	**mes 9**	**mes 10**	**mes 11**	**mes 12**
comp fibra	Compra de fibra de alpaca		13.869,34	7.845,05	4.437,47	2.510,01	15.191,50	16.382,75	13.672,98	10.226,33	20.869,00	23.529,41	23.529,41	23.529,41
	Monto de compra fibra de alpaca		2.496.481,24	1.412.108,98	798.744,94	451.801,87	2.734.469,30	2.948.895,71	2.461.136,34	1.840.738,80	3.756.420,43	4.235.294,12	4.235.294,12	4.235.294,12
	Disponibilidad de compra en Bs		2.496.481,24	1.412.108,98	798.744,94	451.801,87	2.734.469,30	2.948.895,71	2.461.136,34	1.840.738,80	3.756.420,43	4.890.319,53	5.374.338,83	5.364.838,31
0	0	0,00												
Fuentes	top de alpaca		0,00	0,00	0,00	1.985.924,97	1.123.318,06	635.393,33	359.403,71	2.175.242,00	2.345.815,99	1.957.808,46	1.464.288,65	2.988.193,54
	hilo de alpaca		1.131.279,46	639.896,61	361.950,95	204.733,84	1.239.123,65	1.336.290,90	1.115.262,93	834.130,04	1.702.220,39	1.919.221,81	1.919.221,81	1.919.221,89
	total fuentes		1.131.279,46	639.896,61	361.950,95	2.190.658,80	2.362.441,72	1.971.684,22	1.474.666,64	3.009.372,04	4.048.036,38	3.877.030,27	3.383.510,46	4.907.415,43
Saldo			1.131.279,46	639.896,61	361.950,95	2.190.658,80	2.362.441,72	1.971.684,22	1.474.666,64	3.009.372,04	4.048.036,38	4.532.055,68	4.522.555,17	6.036.959,62
Ventas	Venta de tops		2.482.406,21	1.404.147,58	794.241,66	449.254,64	2.719.052,50	2.932.269,99	2.447.260,57	1.830.360,81	3.735.241,92	4.211.415,75	4.211.415,75	4.211.415,69
	Venta de hilos		1.131.279,46	639.896,61	361.950,95	204.733,84	1.239.123,65	1.336.290,90	1.115.262,93	834.130,04	1.702.220,39	1.919.221,81	1.919.221,81	1.919.221,89
	Ventas totales		3.613.685,67	2.044.044,19	1.156.192,60	653.988,47	3.958.176,16	4.268.560,88	3.562.523,51	2.664.490,85	5.437.462,31	6.130.637,55	6.130.637,55	6.130.637,58
	Ganancia		1.117.204,43	631.935,21	357.447,66	202.186,60	1.223.706,85	1.319.665,17	1.101.387,17	823.752,05	1.681.041,88	1.895.343,44	1.895.343,44	1.895.343,46
													Gan totales	**14.144.357,36**
	Participación porcentual top		69%	69%	69%	69%	69%	69%	69%	69%	69%	69%	69%	69%
	participación porcentual hilo		31%	31%	31%	31%	31%	31%	31%	31%	31%	31%	31%	31%
	Riesgo calculado		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
	Riesgo asumible		10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
Uso de capacid	top		59%	33%	19%	11%	65%	70%	58%	43%	89%	100%	100%	100%
	hilo		18%	10%	6%	3%	19%	21%	17%	13%	26%	30%	30%	30%

In addition to the initial availability of resources, this table includes the calculations corresponding to sales revenue. For yarn, this revenue is available in the month in which the sale takes place; for exports, the remaining 80% is available after three months. Therefore, these sources of revenue are used in the following month for the purchase of fiber. The table also includes the resulting amount of sales for the period, the profit obtained by subtracting purchases (in practice, the difference, since fixed expenses are not included), the percentage share of top and yarn sales, and finally the risk calculated for each period, which must not exceed 10%.

Finally, the dialog box corresponding to the solution of the proposed model using Excel Solver for a nonlinear optimization problem is shown.

Figure 3. Application of Excel Solver.



IV. CONCLUSIONS AND RECOMMENDATIONS

- A nonlinear programming model has been applied, and a solution has been obtained. The model may be expanded and adjusted according to variations in the parameters used.
- The model should be applied to the product mix, or adjusted according to the inclusion of new products using the templates generated.
- Excel Solver is an appropriate tool for working with small- and medium-scale optimization models.
- The application of mathematical models to different business situations makes it possible to obtain solutions that improve economic performance.